

# **Renewable Energies and Infrastructure Investments in Africa: The Synergy Between Private Equity and Project Finance**

*A Literature Review*

## **1. Introduction to the Literature**

Africa's energy transition sits at the intersection of three powerful currents in contemporary finance: the search for yield in long-duration real assets, the institutional push for climate-aligned capital allocation, and the persistent challenge of mobilising private money for development. This literature review synthesises the academic and practitioner work that informs how Private Equity (PE) sponsorship interacts with non-recourse project finance to make greenfield renewable energy assets bankable in African markets. The argument running through the chapters that follow is that PE does not simply provide equity tickets; it operates as a development-stage risk buffer whose discipline, governance, and capital structuring choices unlock senior debt that would otherwise remain on the sidelines (Esty, 2004; Steffen, 2018). The literature is rich on each individual building block, including project finance theory, infrastructure as an asset class, African power sector reform, and blended finance, yet the specific mechanism by which PE absorbs early-stage uncertainty to crowd in commercial lenders has been examined only in fragments. This review maps those fragments, identifies the conceptual scaffolding they share, and sets out the gap that motivates a comparative study of North African and Sub-Saharan African markets.

The review proceeds in eight substantive sections. It begins with the macro picture of Africa's infrastructure deficit and the role of renewables. It then turns to the micro-financial machinery of project finance and the concept of bankability. The third strand examines infrastructure-focused PE, first in mature markets and then in African frontier conditions, before considering how PE specifically absorbs greenfield risk. Risk mitigation tools, blended finance architecture, and the ESG to bankability nexus follow. A comparative section contrasts North Africa's more institutionalised PE ecosystem with the Sub-Saharan environment. The review closes by articulating the unresolved research question.

## **2. Africa's Infrastructure Gap and the Role of Renewable Energy Investment**

The scale of Africa's infrastructure deficit is now well-documented. The African Development Bank estimates that the continent requires between USD 130 billion and USD 170 billion annually to meet its infrastructure needs, leaving an unfunded gap of USD 68 to 108 billion every year (AfDB, 2018; AfDB, 2023). The Programme for Infrastructure Development in Africa (PIDA), adopted by the African Union, identifies 69 priority projects with an estimated cost of USD 160.8 billion through 2030, yet implementation has lagged because of weak project preparation pipelines and constrained sovereign balance sheets (UNECA, 2023). Calderón and Servén (2010), in their World Bank work on Sub-Saharan Africa, demonstrated that the infrastructure deficit shaves roughly two percentage points off annual GDP growth and reduces firm-level productivity by up to forty percent. These findings have anchored most subsequent analyses of the cost of inaction.

Within this aggregate gap, electricity is the binding constraint. Eberhard and Shkaratan (2012) showed that Sub-Saharan Africa's installed capacity, excluding South Africa, was roughly equivalent to that of a single mid-sized European country, despite a population approaching one billion. Eberhard, Gratwick, Morella, and Antmann (2017) updated this picture and found that Independent Power Producers (IPPs) had become the fastest-growing source of generation investment on the continent, channelled almost exclusively through limited-recourse project finance structures with long-tenor Power Purchase Agreements (PPAs). Renewable technologies (utility-scale solar PV, onshore wind, and increasingly battery storage) have shifted from niche to mainstream within this IPP wave, partly because Africa's resource endowment is exceptional and partly because levelised costs have fallen sharply (IRENA, 2022; Kruger & Eberhard, 2018).

The role of renewable energy is therefore double-sided. It addresses a development imperative (universal access under SDG 7) while offering a cost-competitive answer to capacity shortfalls. However, Schmidt (2014) demonstrated in *Nature Climate Change* that the cost of capital, rather than the cost of equipment, is the dominant variable shaping the viability of low-carbon investment in non-OECD countries. Because renewables are capital-intensive at the front end and have no fuel cost, every percentage point added to the weighted average cost of capital translates directly into higher tariffs. In African settings, sovereign risk premia, off-taker fragility, and currency mismatch combine to push the cost of capital well above OECD benchmarks (Egli, Steffen, & Schmidt, 2018; Schmidt, Steffen, Egli, Pahle, Tietjen, & Edenhofer, 2019). Closing the gap is therefore not only a question of mobilising more capital but of structuring it more efficiently.

### 3. The Mechanics of Project Finance and the Concept of Bankability

Project finance is the dominant technique for funding utility-scale energy assets in emerging markets. Gatti (2023), in the fourth edition of *Project Finance in Theory and Practice*, defines it as the financing of a discrete economic unit through a Special Purpose Vehicle (SPV), where lenders rely primarily on the cash flows of the project, not the balance sheet of the sponsor, for debt service. This non-recourse architecture allocates each risk (construction, operation, demand, regulatory, force majeure) to the contractual party best able to bear it, and it permits debt-to-equity ratios that would be implausible in a corporate finance context, often 70:30 or higher for renewables (Yescombe, 2014; Gatti, 2023).

The classical theoretical case for project finance was articulated by Brealey, Cooper, and Habib (1996), who argued that segregating cash flows in an SPV reduces information asymmetry and contamination risk between the sponsor and the asset. Esty (2004), in his agenda-setting *European Financial Management* paper, extended this analysis by showing that project finance is selected for assets characterised by tangibility, high free cash flow, and discrete project boundaries, precisely the features of a wind or solar plant. In a related study, Esty (2002) documented that as project finance migrated from low-risk natural-resource deals into emerging-market infrastructure, expected returns rose, default rates increased, and debt capacities tightened. Sawant (2010a, 2010b) formalised this further, demonstrating that high leverage and syndicated debt structures with development bank participation actively reduce the risk of host-government expropriation by raising the political cost of default. Müllner (2017), in his review for *Management Review Quarterly*, synthesised three decades of project finance research and concluded that the technique's strategic value lies in its ability to discipline sponsor behaviour and credibly precommit cash flows, a finding directly relevant to African contexts.

Bankability is the operational test of all this theory. A project is bankable when senior lenders are willing to extend long-tenor non-recourse debt at a Debt Service Coverage Ratio (DSCR) and pricing that allow equity returns to clear sponsor hurdles. Borgonovo, Gatti, and Peccati (2010) showed empirically that the determinants of value creation in project finance transactions cluster around contractual completeness, particularly the strength of the PPA, the credit quality of the off-taker, and the robustness of construction guarantees. Steffen (2018) provided one of the few large-N empirical analyses of project finance for renewables and found that, even in low-risk OECD settings, the technique is used disproportionately for renewable assets because it allows non-utility developers to overcome debt overhang and access cheaper

senior tranches. The implication for African markets is direct: bankability is not a property that emerges spontaneously but is engineered through layered contracting, credit enhancement, and equity sponsorship of a calibre that lenders trust.

#### **4. Private Equity in Infrastructure: From Mature Markets to Emerging Frontiers**

PE infrastructure as a distinct asset class is a relatively recent construct. Inderst (2009) traced its emergence in OECD markets to the search for inflation-linked, long-duration cash flows that match pension fund liabilities. Della Croce and Yermo (2013) and Della Croce and Gatti (2014) documented how unlisted infrastructure funds, typically closed-end vehicles with ten-to-fifteen-year horizons, became the principal channel through which institutional capital reached project equity. Sawant (2010b), in his Wiley monograph *Infrastructure Investing*, argued that infrastructure's economic characteristics (large upfront capex, stable cash flows, regulatory entanglement, exposure to political holdup) make capital structure choices a strategic variable rather than a residual one.

The intellectual lineage of infrastructure PE runs through the leveraged buyout literature. Kaplan and Strömberg (2009) provided the canonical treatment of the PE governance model: concentrated ownership, active board engagement, leverage discipline, and a defined exit horizon. Although their work focused on operational buyouts in mature markets, the governance toolkit they describe has migrated into infrastructure investing. Gupta and Van Nieuwerburgh (2021), in *The Review of Financial Studies*, found that closed-end infrastructure funds in fact deliver risk-return profiles closer to private equity than to the long-duration income story sometimes told to LPs, with performance strongly tied to deal exit speed. This finding qualifies the simple narrative of infrastructure as a stable bond proxy and is important when reading African PE returns, where exit markets are thinner.

In emerging markets, the calculus shifts. Leeds (2015), in *Private Equity Investing in Emerging Markets*, argued that PE plays a fundamentally different role in developing economies than in OECD markets: it substitutes for absent public equity markets, supplies governance and operational expertise, and bears risks that local banks will not. Inderst and Stewart (2014), in their World Bank policy research paper, showed that institutional allocation to emerging-market infrastructure remains a small fraction of the addressable pool, hampered by data scarcity, regulatory fragmentation, and the difficulty of building in-country deal pipelines. Within Africa specifically, the African Private Capital Association (AVCA, 2024, 2025)

reports that infrastructure now accounts for around thirty percent of total private capital fundraising on the continent, with renewables driving the majority of new commitments. Specialised funds such as Actis Energy, African Infrastructure Investment Managers (AIIM), Meridiam, Helios Investment Partners, Inspired Evolution, and the development-led platforms backed by British International Investment (BII), Proparco and FMO have professionalised the sponsor base and increasingly co-invest with multilateral DFIs (AVCA, 2025; Eberhard et al., 2017).

## 5. PE as a De-risking Mechanism in Greenfield Infrastructure

The greenfield phase is where the synergy between PE and project finance becomes most visible. Schmidt (2014) and Egli, Steffen, and Schmidt (2018) showed that the largest cost-of-capital gap between OECD and developing markets arises during development and construction, when revenues are zero and the SPV is at its most fragile. Senior lenders typically demand that this period be substantially de-risked before drawing down debt. Equity sponsors must therefore carry the project through site control, environmental and social impact assessments, grid connection studies, EPC contractor selection, and PPA negotiation, often eighteen to thirty months of expenditure with a real option of project failure (Yescombe, 2014; Gatti, 2023).

This is the developmental work that a credible PE sponsor underwrites. Esty (2002) argued that the equity sponsor's reputational capital, technical capacity, and willingness to absorb sunk development costs are what convert a paper project into a financeable one. In the African context, Eberhard and Gratwick (2011), analysing the determinants of IPP success in Sub-Saharan Africa, found that the presence of an experienced sponsor with a track record and the capacity to absorb early-stage risk was among the strongest predictors of reaching financial close. Their later cross-country study (Eberhard et al., 2017) confirmed that sponsor quality, alongside the credit quality of the off-taker and the strength of the procurement framework, separated successful programmes (South Africa, Kenya, Uganda) from stalled ones.

The economic logic was formalised by Sawant (2010a) in the *Journal of International Business Studies*. He showed that in infrastructure FDI, capital structure is itself a risk-mitigation tool: high leverage signals discipline, syndicated debt with DFI participation creates a reputational deterrent against expropriation, and the SPV form caps sponsor exposure. PE sponsors are unusually well placed to engineer this structure because they can iterate quickly on equity terms, bring in co-investors, and accept a development-phase IRR profile that strategics often

will not. Müllner (2017) reinforced this point: PE-backed SPVs typically display tighter covenant packages and more granular risk allocation than corporate-financed projects, which translates into lower debt margins and longer tenors.

The pattern observable in the data is therefore one of sequential risk transfer. PE equity absorbs development and early-construction risk; once milestones are hit (PPA signature, EPC fixed-price lump-sum contract, lender's technical adviser sign-off), commercial debt comes in alongside DFI senior tranches; and once the asset has demonstrated operational stability, the sponsor refinances or sells to long-duration institutional buyers (Della Croce & Gatti, 2014; AVCA, 2025). This three-stage pipeline is, in effect, the financial symbiosis that this thesis investigates.

## **6. Risk Mitigation Tools and Blended Finance Structures**

Even with a strong PE sponsor, raw African project risk often exceeds the appetite of commercial lenders. The literature on risk mitigation instruments and blended finance describes the layered architecture that fills this gap. Komendantova, Patt, Barras, and Battaglini (2012), in an early empirical study of Concentrated Solar Power in North Africa, showed that political and regulatory risk perceptions, rather than technological risk, dominate investor decision-making. Their work helped frame the case for systematic public-sector de-risking.

The instruments themselves cluster into three families. The first is political risk insurance, principally provided by the Multilateral Investment Guarantee Agency (MIGA) of the World Bank Group and by the African Trade and Investment Development Insurance Agency (ATIDI). These products cover expropriation, currency inconvertibility, transfer restrictions, breach of contract, and war and civil disturbance, and their use in renewable energy IPPs across Côte d'Ivoire, Kenya, Egypt and Uganda has become standard (MIGA, 2024; ATIDI, 2024). The second family consists of partial risk and partial credit guarantees from multilateral lenders, which protect against off-taker non-payment, the dominant residual risk in markets with financially distressed state utilities (Eberhard & Gratwick, 2011; Climate Policy Initiative, 2015). The third family is blended concessional finance, in which DFIs deploy junior tranches, first-loss equity, or below-market debt to crowd in commercial capital.

The academic and practitioner literature on blended finance has expanded considerably. Flammer and Giroux (2024), in an NBER working paper, formalised the decision-making of DFIs that engage in blended finance and provided empirical evidence that concessional capital is concentrated in Africa, particularly for renewables and infrastructure. The IFC's joint DFI

Working Group reports (IFC, 2023, 2025) document that infrastructure absorbed the largest share of blended concessional finance commitments globally and that Sub-Saharan Africa was the dominant regional recipient. AVCA (2025) reports that between 2013 and 2022, Sub-Saharan Africa attracted forty-one percent of the volume and fifty percent of the value of global blended-finance infrastructure deals. Léon (2025), writing in *World Development*, has, however, sounded a cautionary note: in his analysis of intermediated DFI lending programmes in Africa, he found that supported banks reduced loan growth post-programme by roughly eight percent, suggesting that the catalytic claims of blended finance need to be tested empirically rather than assumed.

The Scaling Solar programme led by the IFC, applied first in Zambia and later in Senegal, illustrates the integration of these tools. It bundles standardised tendering, a model PPA, a World Bank IDA guarantee on off-taker payment, MIGA political risk cover, and IFC senior debt into a single template (World Bank, 2020). Kruger and Eberhard (2018) compared Scaling Solar in Zambia with South Africa's REIPPPP and Uganda's GET FiT programme, finding that bundled risk mitigation translated directly into bid prices: tariffs of USD 6.0 to 7.8 per kWh in Zambia were among the lowest in the region at the time. The architecture works precisely because PE sponsors absorb the residual risk that no public instrument covers, typically completion, performance, and resource risk.

## **7. ESG as a Catalyst for Bankability and Value Creation**

ESG integration in emerging-market PE has moved from compliance overhead to a recognised value-creation lever. The Equator Principles, embedded into the lending policies of the International Finance Corporation and the European DFIs, require Environmental and Social Action Plans (ESAPs) as a precondition for senior debt drawdown (Gatti, 2023). For PE sponsors of African renewable assets, this means that credible ESG processes such as biodiversity assessment, community engagement, labour standards, grievance mechanisms, and climate-risk modelling are the operational ticket to debt unlock, not an optional reporting layer.

The literature on ESG and PE returns is still maturing. Zaccone and Pedrini (2020), writing in *Sustainability*, found that ESG integration in PE deal sourcing and portfolio management is positively associated with operational performance, particularly in emerging markets where ESG capacity is itself scarce and therefore commands a premium. Pucker (2021), in *Harvard Business Review*, has been more sceptical, arguing that much of the claimed performance lift

reflects selection effects rather than causal value creation. Greenfield, Monk, and Rook (2026), in their SSRN working paper drawing on British Columbia Investment Management Corporation data, attempted to quantify ESG-related uplift in projected enterprise values and identified a measurable double-digit effect, although their sample is limited.

For African renewables specifically, the ESG to bankability nexus is more direct than the abstract debate suggests. A wind farm without robust avifaunal studies will not pass a lender's technical adviser screen; a solar plant without clear land-tenure documentation will not secure MIGA cover; a mini-grid project without a gender-disaggregated community engagement plan will not access concessional climate finance from the Green Climate Fund or the Sustainable Energy Fund for Africa (AfDB, 2024). PE sponsors that internalise these requirements at the development stage compress the time to financial close and reduce contingent equity commitments. ESG, in this sense, is not a parallel objective to financial structuring; it is a component of bankability.

## **8. Comparative Perspectives: North Africa vs Sub-Saharan Africa**

The literature treats North Africa and Sub-Saharan Africa as distinct investment ecosystems, and the comparison is instructive. Morocco's Plan Solaire and the institutional framework around the Moroccan Agency for Sustainable Energy (MASEN) created what Boulakhbar et al. (2020) and El-Katiri (2016) describe as a centralised, state-led PPP model in which the sovereign carries much of the off-taker risk through its guarantee of MASEN obligations. The Noor Ouarzazate complex, financed through a layered structure involving the African Development Bank, the World Bank, the European Investment Bank, the Clean Technology Fund, and Saudi-led equity sponsors, has become the reference case for utility-scale CSP in the region (AfDB, 2018). Egypt's Benban solar park, developed under a feed-in-tariff round and project-financed by a consortium led by IFC and the EBRD, similarly mobilised more than USD two billion in private capital under sovereign guarantees (IFC, 2019). North African deals therefore tend to be larger, more centralised, and more structured around sovereign credit enhancement.

Sub-Saharan Africa presents a more fragmented but also more dynamic picture. South Africa's REIPPPP, examined extensively by Eberhard and Naude (2016), Baker and Phillips (2019), and Kruger, Stritzke, and Trotter (2019), has procured more than 6,300 MW of renewable capacity across multiple bid windows and has become the most-cited example of a successful auction-based procurement programme in the developing world. Outside South Africa,

however, transactions are smaller, off-takers are weaker, and the mix of capital is more heterogeneous. Eberhard, Gratwick, Morella, and Antmann (2017) document that Kenya, Uganda, Senegal, Côte d'Ivoire and Zambia have produced replicable IPP templates, but each remains dependent on DFI senior debt and political risk cover to reach financial close. The AVCA (2024) data confirms the implication: PE fundraising in Sub-Saharan Africa is increasingly specialised, with dedicated infrastructure funds (AIIM, Meridiam, Inspired Evolution, Helios) accounting for an unprecedented thirty percent of 2024 final-close values, but average deal sizes remain modest.

The comparison therefore points to two distinct PE roles. In North Africa, PE tends to operate alongside large strategic developers (Acwa Power, EDF Renewables, Engie, Masdar) and competes for equity tickets in sovereign-anchored auctions. In Sub-Saharan Africa, PE is more often the platform builder, aggregating projects under a single sponsor, providing the development capital that strategics will not, and creating exit-ready assets for future refinancing. Bayliss and Van Waeyenberge (2018), writing in the *Journal of Development Studies*, characterise this as a financialisation of development, in which PE intermediation reshapes who bears risk and who captures value across the project lifecycle. Their critique is important: it reminds the analyst that the PE and project finance synergy is not value-neutral and that distributional outcomes (tariff levels, local content, fiscal exposure) depend on how the synergy is governed.

## 9. The Research Gap

The literature reviewed above is rich but uneven. Project finance theory has been thoroughly mapped (Esty, 2004; Gatti, 2023; Yescombe, 2014; Müllner, 2017). Infrastructure as an asset class has been studied at the institutional level (Inderst, 2009; Della Croce & Gatti, 2014; Sawant, 2010b; Gupta & Van Nieuwerburgh, 2021). African IPP dynamics have been documented in detail by the Eberhard and Gratwick body of work and by World Bank syntheses (Eberhard et al., 2016, 2017; Kruger & Eberhard, 2018). The cost of capital and de-risking literature, anchored by Schmidt and his ETH Zürich co-authors, has clarified the macro-financial mechanics. Blended finance has its own growing empirical base (Flammer & Giroux, 2024; Léon, 2025; IFC, 2025).

What remains under-explored is the specific causal channel through which PE sponsorship, as a development-stage risk buffer, unlocks commercial debt in the post-2022 African macroeconomic environment. Three features of that environment matter. First, post-COVID

monetary tightening pushed dollar funding costs to levels that have repriced almost every emerging-market project finance deal, reducing sponsor IRRs and stretching DSCRs (Schmidt et al., 2019; AVCA, 2025). Second, multiple African sovereigns (Ghana, Zambia, Ethiopia, and Kenya at the margin) have entered debt distress or restructuring, weakening the sovereign guarantees and off-taker credit that historically anchored bankability (IMF, 2024). Third, currency depreciation across the continent has widened the gap between local-currency revenues under PPAs and dollar-denominated debt service. Together these conditions have created a new test for the PE and project finance symbiosis: can sponsor equity and DFI catalytic capital still mobilise commercial debt at scale when each underlying risk premium has shifted upward?

A second dimension of the gap is comparative. The literature treats North Africa and Sub-Saharan Africa largely as separate research objects, and there is no systematic study of how the more mature North African PE ecosystem, with its centralised auctions, sovereign guarantees, and deeper local capital markets, differs from the emerging Sub-Saharan ecosystem in the way PE absorbs greenfield risk. The available evidence suggests that the *function* of PE differs across the two regions: capital aggregator and co-investor in North Africa, platform builder and risk underwriter in Sub-Saharan Africa. But this distinction has not been formalised, tested against transaction data, or related explicitly to debt mobilisation outcomes.

The thesis that follows takes up these two threads. It investigates how PE sponsorship transforms greenfield African renewable energy projects into bankable propositions under current macro-financial stress, and how this transformation differs between the two regional ecosystems. By treating PE not as a passive capital provider but as an active risk-engineering layer that operates upstream of debt, it aims to bring the literatures on project finance, infrastructure investing, African power sector reform, and blended finance into a single analytical frame relevant to both academic researchers and the practitioners who structure these deals.

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